



Geodemographic analysis and the identification of potential business partnerships enabled by transit smart cards

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ABSTRACT

Smart card automated fare payment systems are being adopted by transit agencies around the world. The data-storage characteristics of smart cards present novel opportunities to enhance transit services. On the one hand, there are fare policies, where smart card holders are given specific rebates on the use of the service based on usage patterns or levels. On the other, there are non-fare policies, for instance if holders receive advantages, such as rebates and offers, from commercial partners. The purpose of this paper is to present a geodemographic framework to identify potential commercial partnerships that could exploit the characteristics of smart cards. The framework is demonstrated using data from Montreal, Canada. Household survey data, specifically trip ends, and business data points are jointly used to determine the exposure of various types of establishments to users of the Montreal Metro network. Spatial analysis of business establishments in the neighborhood of metro stations helps to identify potential commercial partners. The results illustrate the potential of geodemographic analysis to generate intelligence of commercial interest.

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1. Introduction

In the past few years, the financial, physical, and social limits to building increased road capacity in many urban areas have become increasingly apparent. Despite decades of investment in road infrastructure aimed primarily at satisfying a seemingly insatiable appetite for private motorization, congestion, emissions, and accessibility issues continue to plague urban transportation systems (Shay and Khattak, 2010). Concurrently, the perception that transit systems are important items in any form of integrated transport strategy has grown steadily. Research in Europe, for instance, suggests that optimal strategies for most cities examined involve a combination of public transport measures, car use charges, but only low-cost improvements in road capacity (May et al., 2000). In North America, the city of Toronto, Canada, has witnessed a growing interest in transit as a way to address many of the issues that auto-oriented development has at best failed to solve, or at worse has even helped to generate (City of Toronto, 2000, 2001). These are not isolated examples.

Given past trends, the goal of creating public transportation alternatives that are sufficiently attractive and efficient to compete with endemic private mobility faces a number of important challenges. For instance, transit systems using

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current technologies cannot meet the levels of flexibility and convenience afforded by the private automobile. Furthermore, many of the social and environmental costs of automobility, including emission of greenhouse gases that contribute to climate change (e.g. Chester et al., 2010), health impacts (e.g. Samimi et al., 2009),³ and exclusionary social processes (e.g. Páez et al., 2009; Sanchez et al., 2004), tend to be external. This means that many of the costs of automobility are deferred over time and space, and are not always evident. In contrast, and despite their potential to alleviate many of the external costs that make current transportation patterns unsustainable (Association du transport urbain du Québec, 2008; Morency and Sioui, 2010), transit systems usually face immediate bottom lines with clear fiscal implications in the form of periodic (e.g. annual) operating and investment budgets.

As the imperative to improve the transportation situation in many cities becomes stronger, transit operators and governments at various levels are increasingly challenged to find innovative ways to make transit ever more attractive, in order to raise ridership and revenue for their systems. One avenue of clear interest to achieve this goal follows the introduction of smart card fare payment systems in many transit networks around the world (e.g. Cheung, 2004; Iseki et al., 2007; Kim and Ahn, 1998; Nielsen, 1997; Shield and Blythe, 1997). From the perspective of transit authorities, the primary reason for introducing smart cards has been to facilitate payment and revenue management. Given their capacity to store information electronically, smart cards make it easier to implement fare policies, such as specific rebates based on usage patterns or levels. An incidental benefit of such automated fare systems is their potential to motivate and facilitate partnerships with firms and establishments interested in leveraging their exposure to large volumes of transit passengers in general, and smart card holders in particular. In this context, non-fare policies also emerge as an option, for instance if partnerships could be established to offer, on an exclusive basis, additional advantages to smart card holders. Examples of non-fare policies include, among other instruments, rebates, offers, discounts, and/or loyalty points from various commercial partners associated with the transit operator (e.g. Blythe, 2000). While direct evidence of the effect of this type of policies on ridership is not widely available, similar measures, including local and shopper transit services, as well as marketing and passenger information, are known to increase patronage (Currie and Wallis, 2008, Table 4). Smart cards therefore emerge as valuable tools that (1) can contribute to increase the attraction of transit by providing a package of services and benefits available exclusively to smart card holders; and (2) can help to increase revenue for transit operators by way of public–private partnerships of mutual interest.

Before assessing the parameters of any business agreements, transit authorities must be able to identify potential commercial partners, their categories, and their level of exposure to smart card holders. From the perspective of prospective partners, identifying their level of exposure to smart card holders may assist with the design and implementation of business plans. The objective of this paper is to demonstrate a simple approach, based on geodemographic analysis, to explore the potential for developing business partnerships that could help to enhance the attractiveness and convenience of the system for transit users, while helping operators to generate additional revenue. Geodemographic analysis is the study of population demographic attributes based on location, using spatially explicit analytical approaches. As demonstrated by previous research (e.g. Buliung and Morency, 2009; Miller, 1999; Páez and Scott, 2004; Yao and Jiang, 2009), the utility and value of spatial analysis and statistical techniques is now becoming firmly established in transportation and land use applications.

Our approach is first to investigate the locational pattern of commercial establishments around transit nodes, concretely their clustering in the surroundings of transit facilities. Secondly, spatial clustering around transit nodes by type of establishment helps to identify potential commercial partners based on their patterns of proximity to facilities and passengers. Finally, clustering analysis based on detailed geodemographic information is used to link ridership and their levels of access to specific types of establishments. This generates potentially actionable business intelligence that could be used to identify and target specific market segments. We use Montreal's Metro system as an illustrative empirical example. We rely on two sources of data. The first one, Montreal's metropolitan travel survey of 2003, provides information on transit travel patterns for a representative sample of residents. Secondly, we extract a selection of business types from an exhaustive business point database for the city generated by InfoCanada and Environics Analytics, two leading providers of marketing data in Canada. The case study is timely and relevant, as the transit authority in Montreal (*Société de Transport de Montréal*: STM) recently created a subsidiary (*Transgesco*) with the specific mission to develop the commercial potential of the system. A smart card fare payment system was introduced in the Greater Montreal Area in 2008, but its full potential to develop commercial partnerships remains to be explored. For illustrative purposes, the analysis reported in this paper is generic by establishment type; however, as should be clear, it is straightforward to extend the analysis to obtain more detailed information, for instance regarding specific franchises, brands, or demographic segments, as required for actual business applications. The results make a persuasive case for the potential of geodemographic analysis for identifying potential commercial partnerships based on the unique characteristics of smart card systems.

The structure of the paper is as follows. Next section provides the context and background for the analysis. This is followed by a description of the data and methods used in the analysis. The results of the analysis are described in Section 4. We conclude the paper with some discussion of the implications of the analysis, and directions for future research.

³ While hotly debated, the full impact on health is yet to be clearly determined, q.v., Badland and Schofield (2008).

2. Context and background

2.1. Firm location analysis

Firm location has been the object of theoretical studies in economic geography. The foundational theory on firm location is due to Hoover (1948; cited in Hoover and Giarratani (1984)). Development of a theory of firm location hinges on the concepts of elasticity of demand and the cost of transportation. Other things being equal, the price paid by the consumer for a good or a service is determined by the cost of delivery, or alternatively the cost of transportation to the point of sale or service. *Delivered price* under this spatial pricing scheme is composed of the *factory price* (i.e. the price quoted at the point of sale) and the cost of transportation, and depends directly on the distance between the consumer and the firm. For goods and services with elastic demand, the friction of distance implies an inverse relationship between quantity demanded and distance, and ultimately, a *range*, the maximum distance (and implicitly price) that consumers are willing to pay before foregoing consumption.

Firm locational behavior is theorized to be directly related to spatial pricing, with establishments acting as profit-maximizing units that consider their market potential (their access to consumers within their range) and their ability to set a price consistent with their marginal costs, all the while paying attention to spatial competition, which tends to split the market along the lines of equal delivered price. Market-oriented firms display a preference to locate in close proximity to the population they aim to serve. For most urban services, the expectation is that exposure to clientele will tend to dominate locational patterns. Since the main product of transportation systems is accessibility, this is expected to be an important locational force shaping the spatial distribution of firms. Other factors may also have an important influence, including the presence of agglomerative effects resulting from input linkages (Cohen and Paul, 2005). Whether transit facilities act as locational attractors is an empirical question to be determined.

2.2. Smart cards as a transit planning tool

Smart cards are portable tools that store information electronically that can be used to validate fare information and authorize access to a transit network. This makes smart cards less subject to fraud than other forms of payment. They do however, generate the challenge to create systems that are highly secure and that do not interfere, but rather facilitate the performance and productivity of a transit network. As a matter of principle, smart card systems should not bring excessive costs to the operator and should not compromise the privacy of the individuals who travel on the network (Dempsey, 2008).

Many authors agree on the benefits brought by smart cards in transit planning and operations, in addition to the benefits in revenue management. Smart card systems provide the ability to examine user behavior in a better way than revealed preference surveys (Bagchi and White, 2005). They permit more flexible and complex fare structures and inter-network integration of fare policies (Attoh-Okine and Shen, 1995). Thanks to the automatization of payments, these systems reduce the operating costs of the fare collection process and ease the drivers' tasks (Attoh-Okine and Shen, 1995; McDonald, 2003). Incidentally, smart cards also can provide a continuous dataset useful to observe trends and evolution of ridership (Utsunomiya et al., 2006).

Smart cards record transactions for each boarding and databases can be built to include information such as date and time, location of stop by onsite monitors in the case of subways, or on-board GPS equipment for buses, card number, fare type, route number, and direction. Among the potential planning uses of smart card data, the following have been explored in the Canadian context: recording and analysis of general ridership statistics by route/day/week/month (Trépanier et al., 2004); trip end estimation when only boarding stops are known (Trépanier et al., 2007); measuring variability in transit user behavior with data mining techniques (Morency et al., 2007); error detection and correction; missing data imputation (Chapleau and Chu, 2007); visualization and modeling transit trip patterns (Chapleau and Chu, 2007; Chu and Chapleau, 2007); comparison between smart card data statistics and large household survey data results for the same time period (Trépanier et al., 2009b); and identification and location of major trip generators in Gatineau area (Trépanier et al., 2009a). So far, however, exploring other commercial opportunities has not received much attention, despite the potential advantages of such opportunities, as discussed next.

2.3. Rationale for developing business partnerships

With the increasing importance of sustainable development in public discourses, it is no surprise to see that transit and the other alternative modes of transportation are receiving more attention. Making transit more attractive and ensuring a correct level of financing for operation and development are becoming strategic issues in urban areas. This can be achieved by increasing the level of subsidy provided by the various levels of government, by increasing the attractiveness and competitiveness of transit in order to attract more users, or by finding innovative sources of funding.

According to Cambridge Systematics et al. (2009), local and regional funding sources for public transportation include: tax- and fee-based funding; common business, activity, and related sources; revenue streams from projects; new "user" or "market" based sources; and financing mechanisms. Of these funding mechanisms, transit systems have traditionally

relied for the most part on tax- and fee-based approaches. The constant need to confront the direct cost of expansion and operation, however, has contributed to leave “moribund” transit systems (Hess and Lombardi, 2005). On the one hand, public support for funding has tended to fluctuate, whereas on the other fee hikes are usually unpopular and regressive, as they affect mainly those who are most at need for mobility, including low income people in search of jobs (e.g. Yi, 2006). Despite a resurgence, it is not uncommon to find transit systems that face situations of chronic underfunding (Hess and Lombardi, 2005). Statistics published by the Canadian Urban Transit Association (2009) indicate that passenger fares account for about 53% of operating revenues in Canadian transit networks. In Montreal, passenger fares accounted for about 40% of the total revenues at the beginning of the decade (Société de Transport de Montréal, 2001) and 48% in 2008 (Société de Transport de Montréal, 2009). This situation remains problematic, since lack of sufficient resources can compromise the ability of transit services to provide a competitive transport alternative.

In terms of growing the market shares, Currie and Wallis (2008) provide an extensive review of the evidence regarding the impacts of a variety of measures on ridership. Measures that aim at affecting the network itself, such as the addition of new routes, increasing frequency, etc., all require considerable levels of investment and financing. Given the adverse fiscal environment, traditional tax- and fee-based approaches are insufficient. As an alternative to traditional financing approaches, a number of transit authorities are seeking innovative revenue tools. For instance, in 2004, Montreal's transit authority created a subsidiary company, Transgesco, with the objective of forming “partnerships with private companies to ensure the strategic development of [the system's] full commercial potential” (www.stm.info). In addition to network improvements, enhancement can also occur by creating indirect benefits for transit users in general, and transit pass holders in particular. New fare technologies such as smart card systems have opened new opportunities for creating innovative funding sources. Smart cards provide a perfect medium to offer additional benefits to transit subscribers. Programs such as “Duo Auto + Bus”, allowing access to a transit network and car sharing system in Montreal, is an example of a transversal benefit offered to transit users (http://www.communauto.com/duo/stm_ENG.html). In point of fact, one specific concern of Transgesco is developing the commercial potential of smart cards.

Smart card holders in Montreal will be surveyed periodically, as will users of any other transportation mode, as part of the regional travel survey. Hence, strategic data will be available from various sources to estimate the number of travelers using facilities near business places. Clearly, this provides useful market intelligence that transit operators can put on the table at the time business partnerships are developed. Such commercial partnerships are useful to retailers because they help to differentiate themselves from the competition (Deleersnyde et al., 2007). For the transit authority, these partnerships can increase card loyalty and customer retention. Card loyalty appears after a certain time, with incentives in absolute value giving better results (Chiara, 2003). For the city, it reinforces the spatial links between retailers and transit network infrastructures, and thus serves as a planning tool to attract and retain commercial activity (Thomas and Bromley, 2003). Indeed, since the presence of subway facilities has positive influence on retailers around the stations (Castillo-Manzano and Lopez-Valpuesta, 2009), transit authorities could recapture part of these benefits through carefully chosen partnerships.

2.4. Smart card experiences in an international context

Smart card initiatives exist around the world. Some involve fare-based approaches that directly involve transit fare policy, such as rebates on fare, electronic money, balance protection, etc. Other policies do not involve fares directly, but depend on commercial partnerships to offer benefits to transit users, including rebates, fidelity points, multi-use smart card, and so on. Examples of non-fare policies appear in Table 1.

Table 1
Smart-card uses for non-fare policies in a public transit context.

Place	Experiments	Impacts
Brussels, Belgium; Lisbon, Portugal; Konstanz, Germany; Paris, France; Venice, Italy (CNA, 2010)	Multi-usage Calypso smart card	Cost reduction, service improvement, flexible fare policies, and revenue improvement
United Kingdom (Davis, 1999)	Implementation of loyalty programs in drug stores, giving shoppers bonus points equal to 4% of purchases	Smart card holders spend an average of 10% more than other customers
Hong Kong (Meadowcroft, 2005)	Cards allowing some non transport-related transactions, e.g. Coca-Cola vending machines, parking, and public phones. Discounts with some retail participants and transit operators	Card became more convenient for users 86% of trips paid using Octopus card in metro and 60 to 70% in buses
Netherlands (Cheung, 2004)	Card not personalized, and so can be used by more than one individual	Frequency and intensity of letting other people use the card increased from 19 to 28%
Singapore (Smart Card Alliance, 2009)	Publicity in newspaper, television, and posters in residential estates Intuitive system. Gradual implementation with existing and new systems working in parallel. Card is accepted by non-transit participants, like restaurants, movie theaters, schools, libraries, and bowling centers	Generated interest. Limited resistance to change. Gave time to users to make the transition between systems and to transit agency to improve system reliability and performance

3. Data and methods

3.1. Information system

In the Greater Montreal area, a smart card system was implemented in 2008, and has yet to be captured by the household travel survey. In order to assess the potential of smart cards, this research work has been implemented with the latest available large-scale travel household survey data from 2003. These surveys have been conducted quinquennially for almost 40 years in the area and gather data on trips, people and households. The sampling rate is about 5% of the population (165,000 people over 3.6 million inhabitants), and the survey collects information regarding one particular travel day of the fall period for each household in the sample. The 2003 survey contains information about some 70,000 households. From this survey we extracted all trips involving the use of public transit. This includes trips conducted entirely using the public transportation network, or partially in the case of kiss-and-ride and park-and-ride trips.

The transit user population of the Greater Montreal Area is composed of 44% men and 56% women. Most trips are made to work (48%), and other purposes include study (35%), leisure (7%) and shopping (10%). Over 50% of the trips are made by households with annual incomes of less than 40,000\$. About 67% of households are composed of three people or less. The local monthly pass is used for payment in 68% of the trips, compared to 18% for tickets and 11% for regional pass. A very small proportion of trips (about 4%) is paid in cash. Transit users travel essentially on Montreal Island, with a fifth of the trips going to the central business district, and 44% going to the larger Montreal-Center district. Only 8% and 12% travel to the eastern and the western parts of the island. 45% of the trips do not involve any transfers while 35% will transfer one time to the destination. Table 2 provides some key facts on metro passengers, based on 24-h alighting (i.e. transit trip end) data.

A second database is used for the analysis. Based on InfoCanada and Environics Analytics data, a business point dataset identifies business places located in the Greater Montreal area, their number of employees, their annual sales, as well as their location. Over 140,000 places are inventoried. Businesses are classified using the 4-digit Primary Standard Industrial Classification code (SIC); a total of 751 codes are used in the dataset. For the purposes of the analysis below, the codes are re-grouped in three sets of categories. In addition, we also extract a subset of specific businesses, accounting for 18.6% of all businesses. This is done for demonstration purposes. These types of businesses and the numbers of establishments (in parenthesis) are as follows: Restaurants (2907), Beauty shops (630), Grocery stores (2370), Medical doctors (8313), Women clothing stores (827), Schools (845), Daycare (782), Drugstores (4103), Bars (545), Banks (2579), Gas stations (1478) and Video clubs (1131).

3.2. Spatial analysis methods: Clustering analysis

Geodemographic analysis in this research is based on the application of spatial clustering techniques. Several such approaches exist, and in this paper we demonstrate the use of nearest neighbor techniques to characterize the proximity between metro stations/passengers and different types of businesses. Nearest neighbor analysis can be implemented by means of the *F*-function (Bailey and Gatrell, 1995). This is a simple spatial analysis technique that gives the cumulative frequency distribution for the first-order nearest neighbor distances from a point to an event. In the analysis that follows, stations and passengers are points, and the events are business establishments of a given type. The function is calculated as follows for a given distance d :

$$F(d) = \frac{1}{n} \sum_{i=1}^n I(d_i^t \leq d) \quad (1)$$

where n is the number of points and d_i^t is the distance from point i to its first-order nearest neighboring event of type t . $I(\cdot)$ is an indicator function that takes the value of 1 if the argument is true and zero otherwise. Quite simply, the function sums the number of points (stations or passengers) whose first neighboring event (business establishment) is within a pre-defined distance d of the point, and then divides by the total number of points in the analysis.

Table 2
Descriptive statistics on travelers.

	Metro passengers	Transit (all modes) passengers	GMA travelers
Number of travelers per day	674 570	1.203 millions	7.977 millions
Average age	34.76	34.20	37.69
Proportion of less than 25 years old	38.03%	38.25%	27.00%
Proportion of elderly (65 years and older)	4.89%	6.94%	8.98%
Men/Women ratio	0.84	0.79	1.01
Proportion of passengers living in household earning less than \$20,000 per year	21.93%	23.05%	12.29%
Proportion of passengers living in non motorized household	39.50%	39.18%	11.52%
Proportion of passengers living alone	19.19%	18.46%	13.03%

Distance d can be changed and the F -function recalculated; graphically, the function is the plot of the values of $F(d)$ as a function of d . It is easy to see that the function itself is a proportion, and therefore bounded between 0 and 1. A value of 0 indicates that no points have a nearest neighbor of type t within distance d , whereas 1 indicates that all points have a nearest neighbor of type t within distance d . The F -function is a descriptive statistic of spatial clustering, and the steepness of the curve indicates the degree and spatial scale of clustering around the reference points: a steep curve at a short distance d provides evidence that events tend to be located in the neighborhood of points. While $F(d)$ operates on a single scale (i.e. first-order neighbors), curves for other neighbor orders, such as second-, third-, and higher orders, can be superimposed to explore clustering at different scales. Frequency curves for different business types or neighbor orders can be compared graphically, and their differences statistically examined by means of the two-sample Kolmogorov–Smirnov statistic. This statistic is a non-parametric minimum distance estimator that tests whether two data samples come from the same (unspecified) distribution.

Analysis in this study was conducted in MATLAB, but F -functions can be easily implemented in any mathematical software that supports arithmetic and logical operators. Specialized spatial analysis software exists as open source (e.g. Bivand et al., 2008; Buliung and Remmel, 2008; Liu and LeSage, 2010; Rey, 2009), and commercial packages are also available that implement spatial analysis routines.

4. Illustrative application

In this section, we illustrate the application of the proposed geodemographic approach to derive information about the spatial relationships between metro stations, passengers, and business establishments. We do this in the following logical sequence. Firstly, it is important to show that commercial establishments tend to cluster in the proximity of transit stations, more than what would be expected by chance. Failing this, it would be difficult to single out transit stations as locations of interest, since they would not be any different from randomly selected points in terms of their proximity to commercial establishments. Secondly, for a partnership to be of mutual interest, it is first important to show that commercial enterprises stand to benefit from partnering with a transit authority. In concrete terms, this means demonstrating that businesses are indeed exposed to the flow of transit passengers. Thirdly, the transit authority may be interested in identifying potential partners. This task requires a more detailed exploration of the types of businesses that are located in the proximity of stations. Finally, it is important also to demonstrate how analysis of passenger flows can provide valuable insights to inform the business plans of potential partners.

4.1. Clustering of business establishments around metro stations

The first part of our analysis is an exploration of the locational pattern of businesses in relation to metro stations in Montreal. Fig. 1 shows the 68 metro stations in the city used for the analysis. Application of the F -function previously described gives information about the proportion of stations with a commercial establishment at a distance of d or less, and thus describes the degree of concentration of establishments around stations. The results of the analysis are shown in Fig. 2. It can be seen there that a vast majority (approx. 80%) of all metro stations have a business establishment within 100 m, and well over 90% of stations have a business establishment within a distance of 200 m. This suggests a pattern of concentration of commercial activities around metro facilities. However, to put this result in context, we must ask whether commercial establishments are more concentrated in the proximity of metro stations compared to any other set of points. To answer this question, we create simulation envelopes to assess the significance of the empirical function for metro stations.

The following procedure is used to produce simulation envelopes. First, a suitably large (in this case $n = 810$) set of random coordinates that cover the area serviced by the metro network is generated. Then, this set of coordinates is sampled without replacement to obtain a list of 68 (to match the number of metro stations) random locations in the area, which are then used to calculate a simulated F -function. This process of sampling and calculation of the F -function is repeated a suitably large number of times (in this case $r = 99$), and finally the envelopes are obtained by extracting the maximum and minimum values of the simulated functions. The envelopes provide an $r\%$ (i.e. 99%) interval of confidence for the empirical function. An empirical curve contained within the envelopes would indicate that the pattern of concentration of events (i.e. businesses establishments) relative to points (i.e. metro stations) is not different from what would be expected from a set of random points in the study area. However, if the empirical function lies outside of the envelope, this would give evidence of a non-random pattern of concentration. Further, the position of the empirical function with respect to the simulation envelopes would provide information about the type of pattern: if located above the envelopes, the events would be more concentrated around points than what would be expected by chance, whereas a function below the envelopes would indicate less concentration or greater repulsion of events with respect to the points.

Fig. 2 shows that the empirical F -function is above the simulation envelopes. This provides evidence that businesses are more concentrated in the proximity of metro stations than around most sets of randomly selected locations. The pattern of concentration observed is the spontaneous outcome of decision processes by myriad decision makers. Given the non-random pattern, it is reasonable to infer that one of the factors that influence the locational decision by business owners is proximity to metro stations. Therefore, it is reasonable to expect that at least some of these business owners might be interested in further exploiting their locational advantage by developing a partnership with the transit authority. However, since the anal-

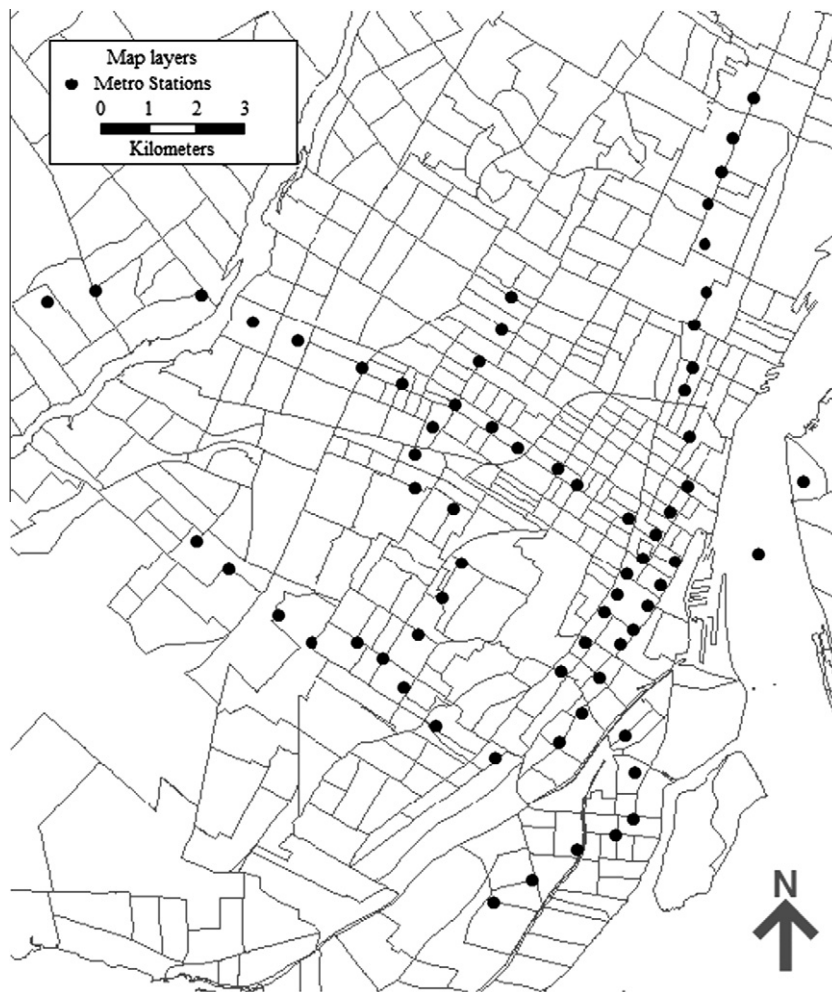


Fig. 1. Metro stations in Montreal.

ysis so far has been only with respect to the concentration of businesses in general, the next step should be to refine the analysis to identify potential commercial partners.

As noted in Section 3.2, the types of business establishments considered in the analysis are: restaurants, beauty shops, grocery stores, medical doctors, women clothing stores, schools, daycare, drugstores, bars, banks, gas stations, and video clubs. The analysis at this point is of the pooled set of business types, and does not yet consider detailed classifications. It bears noting, however, that simulation envelopes can be obtained for any set of events, and therefore significance analysis could be replicated for any business type or combination of businesses as desired.

4.2. Analysis by business type

In this section, we analyze the levels of concentration in a more disaggregate way, in order to identify types of businesses that, given their proximity to stations, could become suitable commercial partners. The results of this analysis appear in Fig. 3, where distinctive patterns by business class can be observed.

Restaurants are the form of business with the highest degree of concentration in the surroundings of metro stations, followed by grocery stores and beauty shops. Notice that approximately 70% of stations have at least one restaurant within 100 m, which is a reasonable walking distance. All stations have at least one restaurant within a distance of less than 450 m. Grocery stores and beauty shops display very similar patterns of concentration up to a distance of about 250 m, at which point they begin to diverge with beauty shops losing concentration more rapidly. The similarity in patterns up to 250 m does not imply that grocery stores and beauty shops tend to co-locate, just that when they first occur they are frequently at more or less the same distance of metro stations. The similarity or dissimilarity of concentration patterns can be explored (see below) by means of statistical testing of the difference of distributions. Co-location patterns are not further

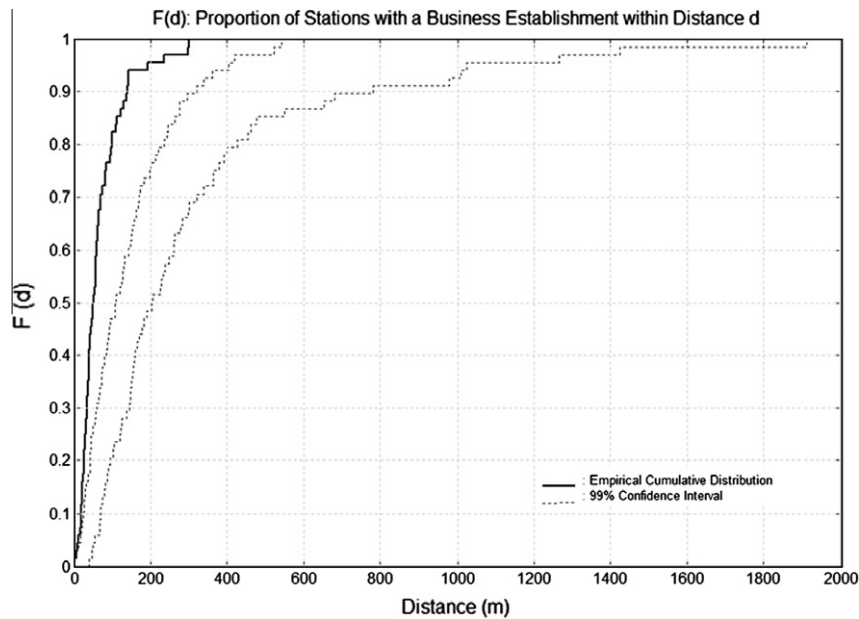


Fig. 2. Spatial clustering of establishments (general) and simulation envelopes.

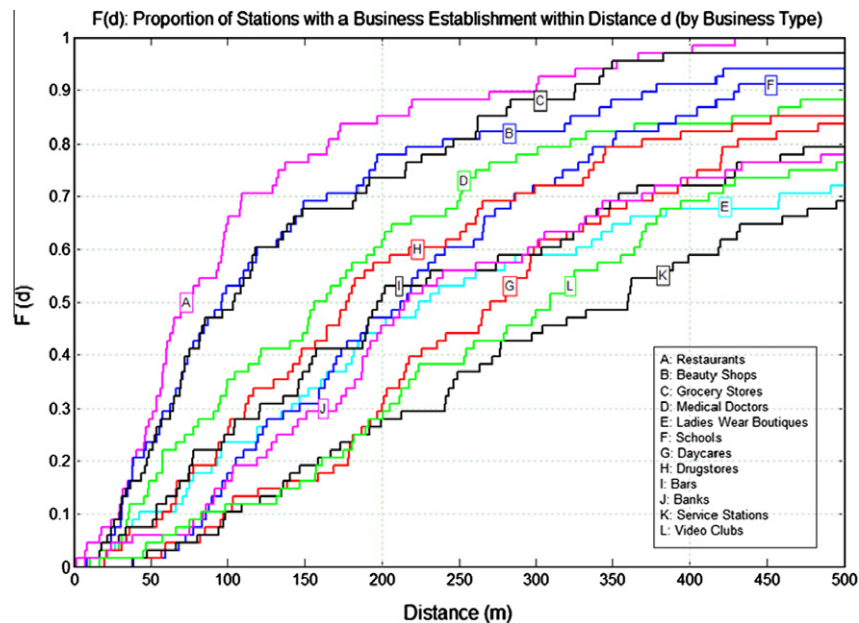


Fig. 3. Spatial clustering of establishments: Proportion of metro stations that have a first-order nearest neighbor of business type t within distance d .

explored here, but we suggest that their study could provide valuable information if the objective was to form “alliances” of businesses based on geographical affinity.

After the three most concentrated types of businesses, there is a wide range of variation of decreasing levels of concentration around stations. At the bottom of the figure, with very similar patterns up to a distance of about 200 m, are daycare facilities, video clubs, and service stations. These establishments have, of all those analyzed, the lowest degrees of concentration around stations. It is possible that daycares and video clubs tend to cater more to a clientele of people who walk to avail themselves of the services offered, or in the case of daycare, perhaps drive to reach them. If this is the case, proximity to stations would represent a marginal advantage at best. It is also possible that these businesses are not sufficiently rentable to command more accessible locations. Service stations, by their very nature, constitute a class of business that clearly targets people with cars, and therefore it is unsurprising that location relative to metro stations is considerably less relevant com-

Table 3

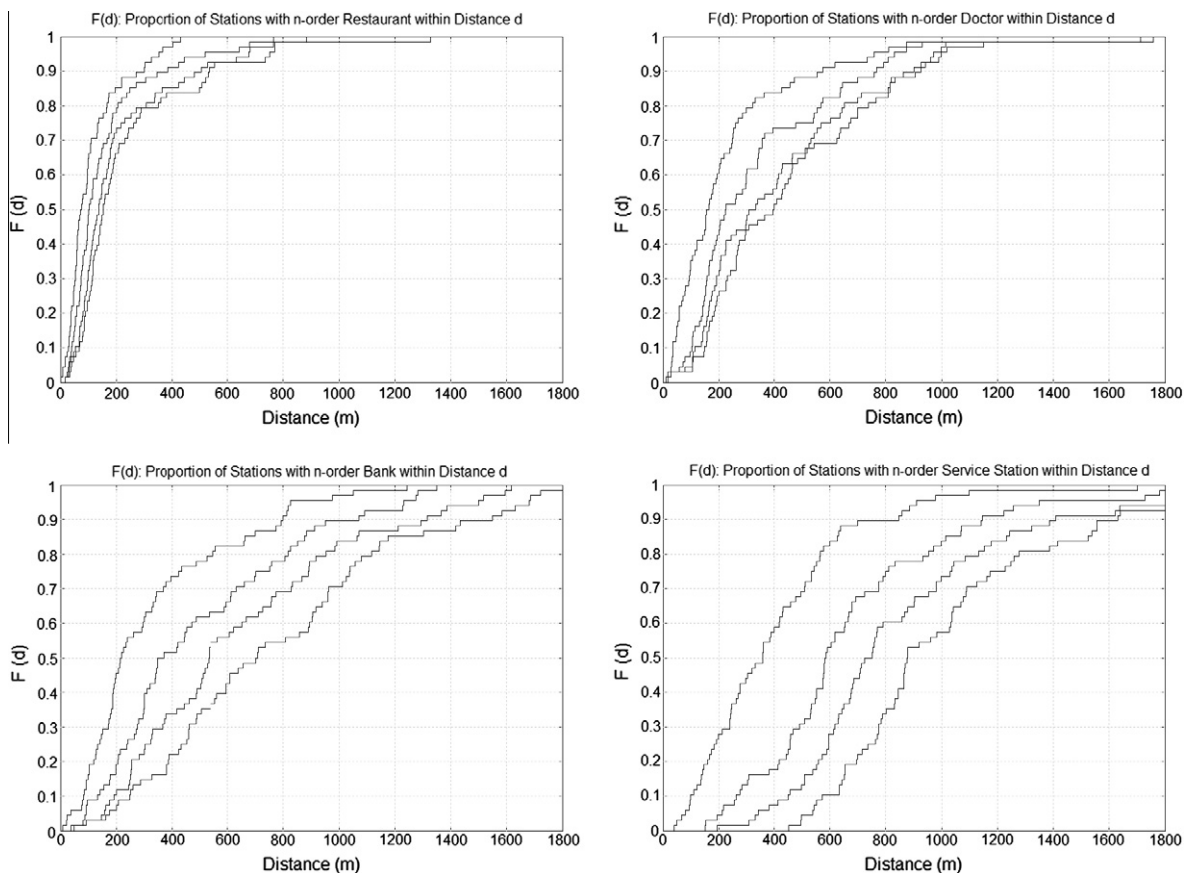
Significance testing, difference of distributions.

	Restaurant	Beauty	Grocery	Doctor	Boutiques	Schools	Daycare	Drugstores	Bars	Banks	Service stations	Videoclubs
Restaurant	X	0	0	1	1	1	1	1	1	1	1	1
Beauty	0	X	0	1	1	1	1	1	1	1	1	1
Grocery	0	0	X	1	1	1	1	1	1	1	1	1
Doctor	1	1	1	X	0	0	1	0	0	1	1	1
Boutiques	1	1	1	0	X	1	0	0	0	0	1	0
Schools	1	1	1	0	1	X	1	0	0	0	1	1
Daycare	1	1	1	1	0	1	X	1	1	0	0	0
Drugstores	1	1	1	0	0	0	1	X	0	0	1	1
Bars	1	1	1	0	0	0	1	0	X	0	1	1
Banks	1	1	1	1	0	0	0	0	0	X	1	0
Service stations	1	1	1	1	1	1	0	1	1	1	X	0
Videoclubs	1	1	1	1	0	1	0	1	1	0	0	X

Note: A value of zero means that the two distributions are NOT significantly different (@ $p = 0.05$).

pared to other business types. Analysis of this function by means of simulation envelopes could be used to determine if, in fact, a pattern of repulsion with metro facilities exists.

As previously noted, differences between the empirical functions can be tested by means of the two-sample Kolmogorov–Smirnov statistic. We conduct an exhaustive pair-wise comparison of all the distributions to determine whether any two in a pair are significantly different from each other. The null hypothesis for the Kolmogorov–Smirnov test is that the values in the empirical functions come from the same continuous distribution, or in other words, that they are statistically undistinguishable. The results are shown in Table 3, where a value of 1 means that the null hypothesis was rejected for the corresponding pair of establishments, and therefore their F -functions (i.e. patterns of concentration) are significantly different. The tests were conducted using a 5% level of significance.

**Fig. 4.** Spatial clustering of selected establishment types (higher order neighbors).

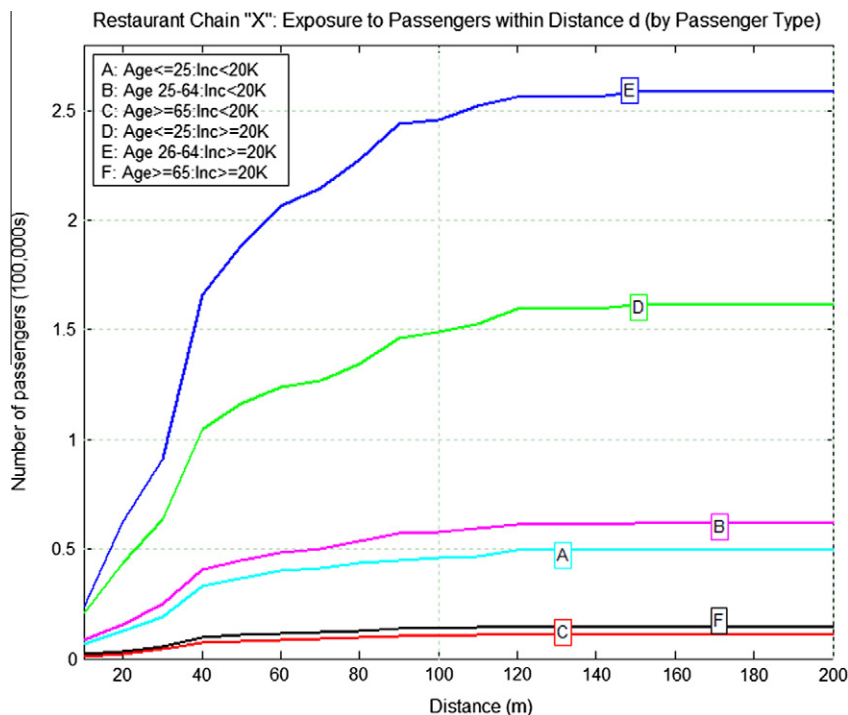


Fig. 5. Exposure of establishment of restaurant chain "X" to different passenger types by distance.

We can see from the table that there are "families" of businesses that display similar spatial patterns of concentration. Two such "families" stand out. Visual inspection of Fig. 2 already suggested that restaurants, grocery stores, and beauty shops were very similar in terms of their location relative to metro stations. Statistical testing confirms that, in fact, their *F*-functions are for all intents and purposes identical. Likewise, service stations, daycares and video clubs form a second clear group of spatially related businesses. Relationships between other business classes could be examined in terms of degrees of belongingness to various families, perhaps by means of fuzzy concepts. For the purpose of illustrating our approach, we select four types of unrelated businesses (in italics in the table): restaurants, doctors, banks, and service stations. It is straightforward to verify that the *F*-functions corresponding to these businesses are all significantly different from each other, and thus can be seen as "representatives" of different spatial "families".

In Fig. 4 we further explore the spatial distribution of restaurants, doctors, banks and service stations, now considering all neighbors up to the fourth degree of proximity. In the figures, the lines represent the functions for the 1st through the 4th nearest neighbor of the corresponding type. Previous results showed that one restaurant can often be found in the vicinity of a metro station. As seen in the first panel of Fig. 4, other restaurants (up to the fourth order of proximity) also tend to locate very near stations: approximately 60% of metro stations have at least 4 restaurants within 200 m. In other words, the spatial distribution of this type of establishment is not only concentrated, it is also compact. This feature begins to vanish for the case of doctors, a class of establishment that we see tends to interpose greater distance between first- and second-order neighbors. Banks and service stations already are characterized by a greater dispersion of establishments, even if the first order neighbor is still relatively close to a large number of stations in the case of banks. For instance, less than 5% of metro stations have more than one service station within 200 m. This information can help to more efficiently target negotiation efforts and agreements.

4.3. Analysis by passenger type

The previous sections demonstrate that businesses tend to locate closer to metro stations than what would be expected by chance. Furthermore, the patterns of concentration vary by establishment type, with some (e.g. restaurants) being more concentrated and having more compact distributions than others (e.g. service stations). While this information can help a transit authority to identify potential partners for developing commercial programs, more is needed to assist partners if they are to decide whether to enter such partnerships. In this section, we illustrate an analysis of business exposure to different demographic segments. To this end, we select from the business database one well-known hamburger chain, H^{***}. The analysis is conducted using a modification of the *F*-function that weights stations based on the number of passengers of a certain category and reports the absolute value as opposed to the proportion. Passenger information is extracted from Montreal's travel survey database. We consider three age groups (less than 25, between 25 and 65, and 65 years and older) as well

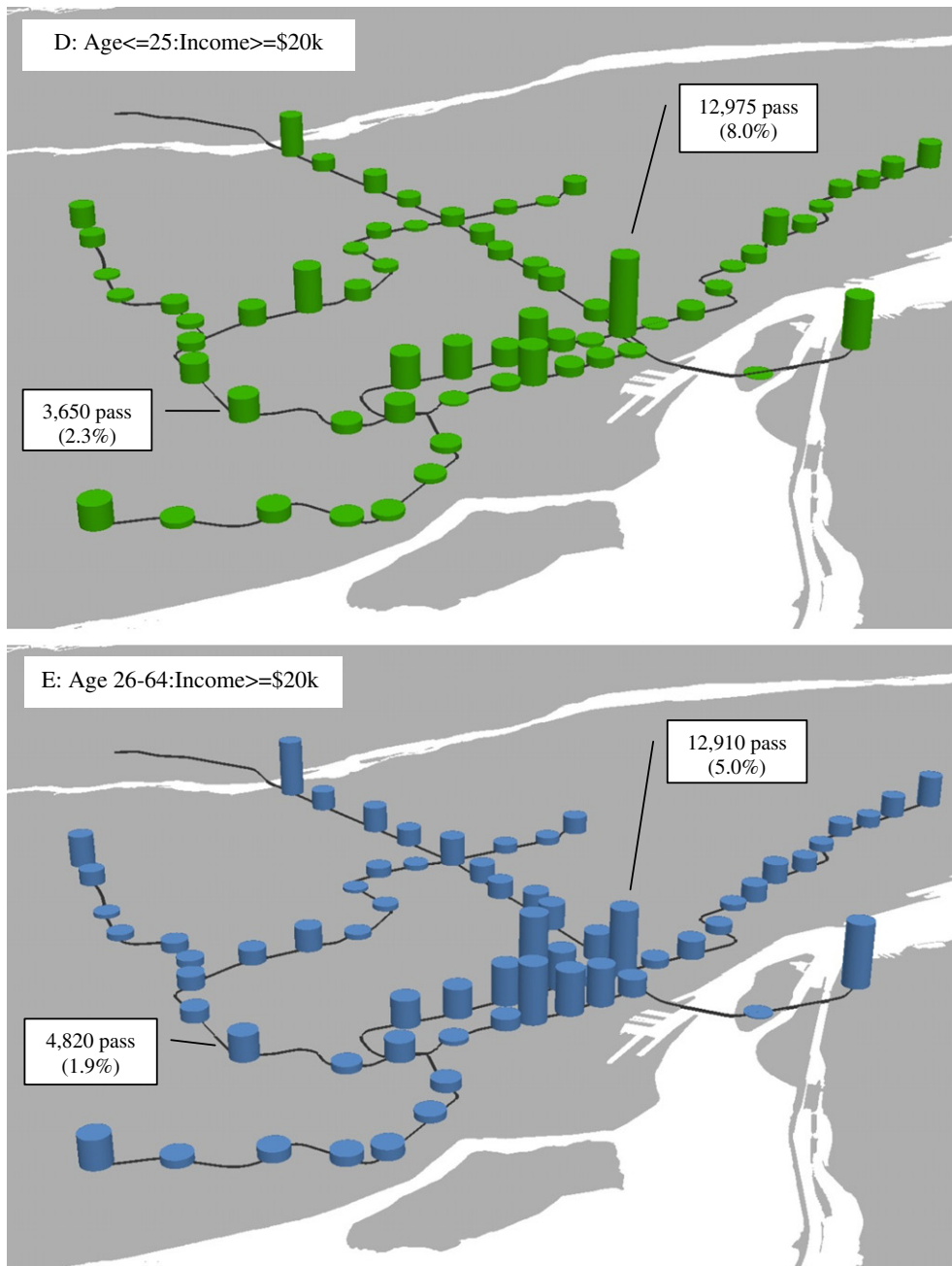


Fig. 6. Geodemographic distribution of passengers (by station).

as two income classes (below and above CAD 20,000). The specific values are not relevant, as the objective is to illustrate the analysis. Other classifications could be created to suit specific analysis needs based on information contained in the household travel survey, including occupation, gender, student status, etc.

The results of the analysis appear in Fig. 5. The proportional patterns of concentration of the four demographic groups in the analysis are similar. As seen in the figure, however, there are important differences in volume (i.e. absolute values). The figure can be interpreted as exposure of establishments of chain H*** to passengers of a certain type. For instance, overall this chain has, within a distance of about 30 m, daily exposure to approximately 100,000 passengers aged 25–64, with incomes greater than 20,000. Approximately 245,000 of these passengers have a restaurant within 100 m of their alighting station. In contrast, there is a smaller demographic of older adults, which reflects a more limited use of transit by this population cohort: less than 30,000 older travelers of both income classes have a restaurant at less than 200 m of their alighting station. While number of older people does not change much according to their level of income, there are substantial differences in

volume when income is considered for other age groups. For example, there are approximately five times more passengers in the 26–64 demographic with household incomes of CAD 20,000 or greater, than there are with household incomes of less than CAD 20,000.

The way various travelers use the system also displays important spatial patterns. Consider two of the demographic profiles above, lower income travelers younger than 25, and travelers aged 26–64 with higher incomes (see Fig. 6). Young, lower income individuals tend to travel to a few well identified stations, two of them associated with institutions of higher education in the Montreal area. Higher income travelers in the 26–64 cohort, in contrast, distribute their much higher volume of traffic over a much wider area of the central part of Montreal. Information of this type can prove useful to develop business plans and to inform business decisions by providing data and benchmarks on traffic volumes, locations, and general concentration patterns of specific industry classes or brands. This can help to address questions such as: where is the best location to get exposure to a certain demographic? For commercial chains, what is the locational pattern of branches with respect to the general level of concentration of other industry actors? And so on.

5. Concluding remarks

In this paper we proposed the use of geodemographic analysis approaches to identify potential commercial partners that could benefit from the introduction of transit smart cards. Geodemographic analysis in this research was based on relatively simple and widely available spatial analysis techniques. The elements of our approach include as proof of principle, a verification of the concentration of business establishments in the neighborhood of transit facilities, a detailed investigation of the types of business that tend to locate near transit facilities, and a spatial analysis of traffic volumes from the perspective of different demographic segments. We illustrated our approach using the case of the metro network in Montreal. Data were obtained from Montreal's household travel survey and complemented with records extracted from an exhaustive business point database. The analysis reported here was purposively kept generic; higher degrees of specificity can be accommodated in a straightforward fashion to attend to the specific needs of actual practical applications.

At a general level, we find that various classes of businesses can be identified as having greater or lesser degrees of concentration around metro stations. The results of the analysis help to discriminate between various establishment types based on their proximity at various degrees. Restaurants, for instance, display patterns of clustering; furthermore, they also tend to be more compactly distributed around stations. Some businesses can further be grouped into “families” of business classes with different degrees of spatial affinity. This includes restaurants, grocery stores, and beauty shops. This information can be used by the transit authority to identify potential partners, and to target efforts to develop partnerships, including coordination of benefits. Given the ability of smart cards to store information, various programs could be implemented: for instance discounts/rebates/loyalty points if customers shop and dine on a transit trip. From the perspective of businesses, analysis of the traveler's data can illuminate levels of exposure to travelers, at different spatial separations from metro stations. Commercial concerns and transit authorities can use this information to explore mutually beneficial arrangements that could exploit the potential of smart cards to implement non-fare policies. These policies hold the potential, as well, to make the use of transit a more attractive, convenient alternative for travelers.

The approach presented in this paper concerns the distribution of commercial activities around transit stations, and the number of passengers at each station. While the analysis makes reference to exposure to travelers by distance, it does not indicate the willingness of travelers to displace themselves a certain distance from the station to reach various types of opportunities. Analysis of the travel behavior of passengers from their destination station could provide valuable complementary information to the one presented in this paper. We suggest that a spatial analysis of such travel behavior could be conducted along the lines of the work of Morency et al. (2011). This is a matter for future research.

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